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Unveiling the frequency-dependent dynamics of acoustic radiation force on lipid-coated ultrasound bubbles

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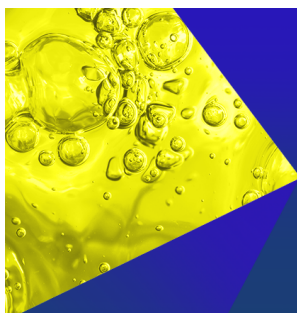
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ABSTRACT

Ultrasonic contrast agent bubbles are a good medium for particle manipulation due to their high acoustic contrast factor. As a kind of widely used acoustic contrast agent, the surface tension of lipid bubbles is complicated, so the solution of its oscillation is normally obtained numerically rather than analytically. Another issue is that the Gorkov's formula for acoustic radiation force (ARF) used for the manipulation of small particles do not apply to bubbles in some frequency ranges. In this paper, the analytical solution of the oscillation of lipid-coated bubbles excited by ultrasound of different frequencies is given, and the frequency dependent ARF of lipid bubbles is obtained theoretically and verified by experiments. The theory complements the widely used Gorkov force, addressing the limitation that Gorkov force is only applicable when the particle density and compressibility are similar to the medium. The results have theoretical significance for the biomedical application of lipid bubbles and may provide theoretical basis for particle manipulation, particularly for the emerging field of manipulating biological particles containing gas bubbles.

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NOMENCLATURE

a_1	Fourier coefficient of x at ω_0 Eq. (49) (-)	T	Period of driving wave, $2\pi/\omega$ (s)
c	Sound speed in liquid (m s^{-1})	t, z	Time and spatial coordinate (s, m)
$\langle F \rangle$	Time-averaged acoustic radiation force (N)	$V(z, t)$	Bubble volume, $4\pi R^3/3$ (m^3)
k	Acoustic wavenumber (m^{-1})	$x(z, t)$	Dimensionless radius perturbation $R = R_0(1 + x)$ (-)
p_a	Amplitude of incident acoustic pressure (Pa)	α	Damping coefficient Eq. (11) (s^{-1})
$p_{ac}(z, t)$	Acoustic pressure field, $p_a \sin kz \cos \omega t$ (Pa)	κ_s	Surface dilatational viscosity (N s m^{-1})
p_0	Ambient (static) pressure (Pa)	μ	Liquid viscosity (Pa s)
R_0	Equilibrium bubble radius (m)	ζ_0	Excitation term amplitude, $-\frac{p_a}{\rho_l R_0^2}$ (s^{-2})
$R_{\text{break-up}}$	Break-up radius threshold (m)	ρ_l	Density of surrounding liquid (kg m^{-3})
R_{buckling}	Buckling radius threshold (m)	$\sigma(R)$	Surface tension [piecewise Eq. (2)] (N m^{-1})
$R(z, t)$	Instantaneous bubble radius (m)	χ	Elastic modulus of bubble shell (N m^{-1})
T_0	Period of $x \cos(\omega_0 t)$, $2\pi/\omega_0$ (s)	Ω_1, Ω_2	Damped natural frequencies (rad s^{-1})
		ω	Angular frequency of excitation (rad s^{-1})
		ω_1, ω_2	Intrinsic angular frequencies Eqs. (9) and (10) (rad s^{-1})

I. INTRODUCTION

Acoustic radiation force (ARF) has been widely used in the manipulation of small particles in the field of medical ultrasound.^{1–7} Bubbles, as the object of ARF manipulation whose acoustic impedance quite different from that of human tissue or water, have gradually come into the researchers' perspective. By introducing programmable phase shift into the standing surface acoustic wave, bubbles can be controlled and shifted to any position in the plane.⁸ Goyal *et al.*⁹ amplified the scattering forces between bubbles by patterning them into arrays and successfully manipulated macroscopic objects at the centimeter level. Jooss *et al.*¹⁰ performed micrometer-scale selective manipulation of microbubbles in the vessels of zebrafish embryos. Gene editing technology has been applied to *Escherichia coli* to generate gas vesicles with protein shells.¹¹ Yang *et al.*¹² utilized the sound sensitivity of this gas vehicle and applied ARF to it by phased-array-based acoustic tweezers, realizing on-demand clustering and movement of these bacteria in live mice.

Since the classic Rayleigh–Plesset equation was proposed,¹³ the bubble dynamics model has been continuously improved and supplemented.^{14–17} Although the small vibrations of free bubbles can be linearized even after taking into account the surface tension at the water–air interface, not all types of bubbles can be analytically solved for their vibrations. In particular, for the widely used lipid-coated bubbles, whose coating can be described by the Marmottant model,¹⁸ the analytical solution of its oscillation can hardly be obtained, because of its piecewise surface tension function. As a result, many numerical and experimental studies of lipid bubbles were conducted to reveal the properties of these bubbles for further application in medical ultrasound.^{19–23} The application of coated bubbles in microfluidic and *in vivo* manipulation has also seen significant advancements in recent years.^{24,25}

As early as 1968, Eller calculated the force on a bubble in a standing acoustic wave by linearizing Rayleigh–Plesset equation for free bubble oscillation.²⁶ Lee and Wang took a different approach to solve the same problem using traditional acoustic methods,²⁷ and the ARF calculated was essentially the same as Eller's. They both revealed that both the magnitude and direction of the ARF exerted on a free bubble change with frequency. However, most of the current research on the manipulation of lipid bubbles use another simpler formula for the ARF: the Gorkov Force.²⁸ According to Gorkov's formula, apart from the external acoustic pressure, the force on a small particle is only related to the density ratio and the elasticity ratio (or the sound speed ratio which is equivalent) between the particle and the medium, and does not include any frequency-related terms which originate from the particle itself. In the detailed derivation of this formula, it can be seen that the scattered acoustic pressure of a tiny particle is directly assumed to involve a monopole term and a dipole term, both independent from frequency.^{29,30} This assumption is true when the density and sound speed of particles are not much different from those of the medium, but it is not accurate for particles such as bubbles, whose sound speed and density are much smaller than water. Therefore, in recent years, there has been growing interest and attention in revising Gorkov's theory.^{31–33} Bjerknes force is a formula for the forces exerted on any particles far smaller than the wavelength, which is suitable for bubbles.^{34,35} However, to get an analytical result, it is necessary to obtain the change of particle volume with time, which is difficult for a lipid bubble. As a result, existing researches mainly focus on the analytical

solution of ARFs on free bubbles with no coating,^{36–38} or merely numerical simulations for lipid bubbles.³⁹

Here, we propose an analytical scheme to obtain the oscillation of lipid bubbles, which can calculate the radius curve near the resonant frequency, and obtain the variation trend under low-frequency or high-frequency excitation. Combining this result with the Bjerknes force gives a more accurate result of the ARF than Gorkov's formula. The experimental results show that bubbles gather at the node of the standing acoustic wave under high frequency while gather at the anti-node under low frequency, proving the correctness of the new theory, making it a supplement to Gorkov's formula.

II. THEORY

Before the theoretical and numerical analysis, Nomenclature summarizes the definitions of all key variables and parameters for the convenience of the reader.

The basic model of the lipid bubbles to be solved is the bubble oscillation equation used to describe lipid bubbles proposed by Marmottant *et al.*,¹⁸ namely

$$\rho_l \left(R\ddot{R} + \frac{3}{2}\dot{R}^2 \right) = \left[p_0 + \frac{2\sigma(R_0)}{R_0} \right] \left(\frac{R}{R_0} \right)^{-3\kappa} \left(1 - \frac{3\kappa}{c} \dot{R} \right) - p_0 - \frac{2\sigma(R)}{R} - \frac{4\mu\dot{R}}{R} - \frac{4\kappa_s\dot{R}}{R^2} - p_{ac}(t), \quad (1)$$

where $\sigma(R)$ is the piecewise surface tension function, described as

$$\sigma(R) = \begin{cases} 0, & R \leq R_{buckling} \\ \chi \left(\frac{R^2}{R_{buckling}^2} - 1 \right), & R_{buckling} \leq R \leq R_{break-up} \\ \sigma_{water}, & \text{ruptured and } R \geq R_{break-up}. \end{cases} \quad (2)$$

In these two equations, R , R_0 , ρ_l , c , p_0 , μ , κ_s , χ , $\sigma(R_0)$, σ_{water} , $R_{break-up}$, $R_{buckling}$ are the changing bubble radius, the initial bubble radius, the liquid density, the liquid sound speed, the ambient pressure, the surface tension when $R = R_0$, the viscosity of the surrounding liquid, the surface dilatational viscosity from the monolayer, the elastic modulus of bubble coat, the surface tension of water, the critical breakup radius of the bubble and the critical buckling radius which is usually close to R_0 , respectively. $p_{ac}(t) = p_a \cos \omega t$ is a harmonic incident acoustic wave, where p_a is its amplitude. This is a widely used model numerically and experimentally verified by researchers, and has been proved to be effective for ultrasound agent bubbles with lipid coating.^{21,40–44}

In the following subsections, first, the vibration characteristics of lipid bubbles excited by different frequencies will be analyzed numerically, so that different methods for analytical solutions fitting these characteristics could be applied. Second, the analytical results of bubble oscillation will be derived under three different situations. Finally, by applying the derived formulas into the Bjerknes force, the ARF on lipid bubbles will be obtained.

A. Characteristics of nonlinear vibration of lipid-coated bubbles at different frequencies

In order to visually get the vibration characteristics of lipid bubbles, a series of numerical investigation are conducted by utilizing the

fourth-order Runge–Kutta method to solve Eq. (1).⁴⁵ A phospholipid bubble with a typical radius of $1.5\ \mu\text{m}$, its resonant frequency estimated to be 3 MHz (the precise value of which will be given in the formulas derivation part), is chosen to calculate its radius curve $R(t)$. The parameters used in the numerical simulation are shown in Table I, based on Refs. 18, 21, 42, and 44. The time step of the numerical simulation is set as 100 000 points in one cycle of the bubble oscillation, i.e., $h = T/100\ 000$, which corresponds to $h = 3.4483 \times 10^{-12}$ in this benchmark problem, but changes as the excitation frequency. Since the fourth-order Runge–Kutta method, as an explicit solution, normally requires a time step $h < T_{\min}/100$, where $T_{\min}$ is the period corresponding to the maximum frequency of change, our selection $h = T/100\ 000$ is sufficient for solving the bubble oscillation problem, even when the oscillation contains high-frequency harmonics.

The radius vibration curve under the excitation of 3.0 MHz incident wave is shown in Fig. 1, where $p_{\text{ext}} = p_{\text{ac}} + p_0$ is the total pressure applied to the bubble, composed of the ambient pressure p_0 and incident acoustic pressure p_{ac} . It can be seen that the oscillation of bubble near the resonant frequency is regular. After being excited by the incident wave, the bubble first goes through a few cycles of transient response, then starts steady vibration, where its amplitude no more changes. The lipid-coated bubble stays in the compression phase during most of the time in a cycle, with its radius smaller than the original radius R_0 , which can be referred to as the “buckled state.” However, the bubble only spends a small part of a cycle in the expansion phase, and, the radius compression amplitude is much larger than the expansion amplitude, which can be referred to as the “elastic state.” This is the “compression-only” behavior of lipid-coated bubbles resulted from their unique surface tension properties.¹⁸ Because the vibration of bubble radius in this case is more regular, it is easy to solve it analytically.

When the frequency of the incident wave is set lower than the resonant frequency, namely 1.0 MHz, 0.6 MHz and 0.3 MHz, a different vibration pattern of bubble radius is observed (see Fig. 2). As in the calculation presented in Fig. 1, the phospholipid-coated bubble in Fig. 2 is also chosen to be $1.5\ \mu\text{m}$ in diameter, with its resonance frequency estimated to be 3 MHz (the precise value of which will be given in the formulas derivation part). Here, we choose three frequencies—1.0, 0.6, and 0.3 MHz—for calculation to provide representative cases where the incident wave frequency is below the resonance frequency. Similar results can be obtained with other sub-resonant incident frequencies as well. When driven at low frequency, the bubble

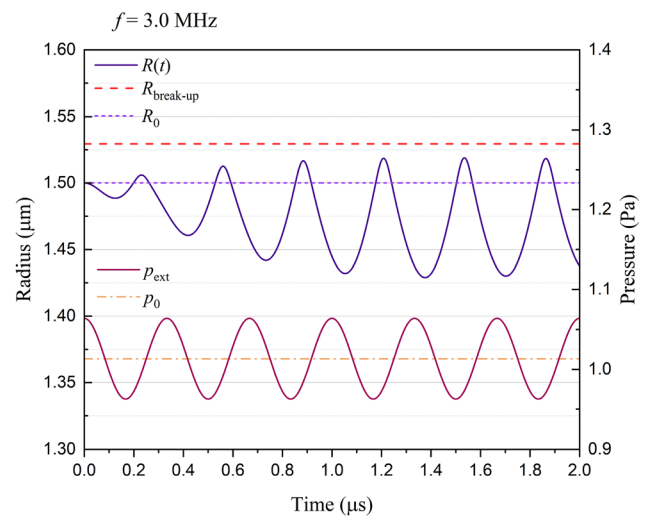


FIG. 1. The vibration curve of the lipid coated bubble near the resonant frequency of 3.0 MHz.

cannot be said to have regular vibration patterns, but at least some of the characteristics of vibration can be summarized. For example, during the compression phase of bubble radius when the acoustic pressure p_{ac} is positive, the bubble rapidly shrinks to a minimum radius and then expands back to the initial radius. While in the expansion phase (when the driving acoustic pressure is negative), the bubble does not grow continuously like a free bubble,¹⁷ but slightly oscillates several cycles at a certain natural frequency because of the existence of the bubble surface tension, which is 3 cycles, 6 cycles, and 11 cycles, respectively, in Figs. 2(a)–2(c).

Although the many small oscillations of the bubble in the expansion phase seem to follow a regular pattern, it does not correspond to the resonant frequency of the bubble, but slightly larger than the resonant frequency (approximately 1–2 times larger). In fact, the natural frequency of this small fluctuation is not only affected by the elastic modulus χ of the bubble coating and the initial radius R_0 of the bubble, but also by the amplitude p_a of the incident acoustic wave. Let us suggest such a scenario: in the half cycle when the incident acoustic pressure is negative, it is as if the bubble is put in an airtight box, where the air inside the box is being slowly pumped out, making the ambient pressure smaller than its original value p_0 . At this time, the equilibrium radius of the bubble is no longer R_0 , but slightly larger. Therefore, the lipid-coated bubble oscillates with a small amplitude in the elastic state when the incident acoustic pressure is negative.

By intercepting one cycle from Figs. 2(b) and 2(c) and enlarging the images, we get Fig. 3. It is obvious that the small fluctuations of bubble radius are mostly above the dotted line which represents R_0 . To obtain the natural frequency of such small fluctuations, we can simply use the formula of intrinsic frequency in the “elastic region” that Marmottant deduced in his article,¹⁸ written as $f = \frac{1}{2\pi} \sqrt{\frac{(3\kappa p'_0 + 4\chi/R)}{\rho_l R_0^3}}$. Here, p'_0 should have been the ambient pressure p_0 , but in this scenario, it is the equivalent ambient pressure after pumping out some air from the box. Although the reduced p'_0 should make the numerator of the formula smaller, the “ $4\chi/R$ ” term plays a more important role,

TABLE I. Bubble parameters in the numerical simulation.

Parameters	Values
c	$1.48 \times 10^3\ \text{m/s}$
ρ_l	$1.00 \times 10^3\ \text{kg/m}^3$
μ	$1.00 \times 10^{-3}\ \text{Pa} \cdot \text{s}$
σ_{water}	$7.29 \times 10^{-2}\ \text{N/m}$
$\sigma(R_0)$	$1.00 \times 10^{-3}\ \text{N/m}$
σ_{break}	$1.10\sigma_{\text{water}}$
χ	$2.00\ \text{N/m}$
κ_s	$5.00 \times 10^{-10}\ \text{N}$
p_0	$1.01 \times 10^5\ \text{Pa}$
κ	1.095

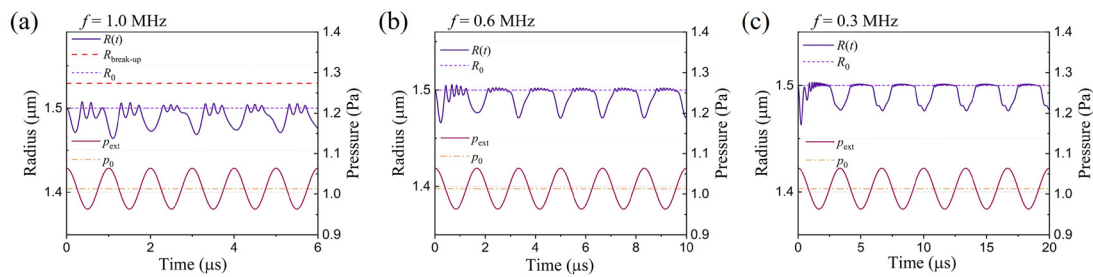


FIG. 2. The vibration curve of the lipid coated bubble under low-frequency excitation. The frequencies are (a) 1.0, (b) 0.6, and (c) 0.3 MHz.

who has a magnitude of 10^6 , larger than p'_0 with merely a magnitude of 10^5 . As a result, for the purpose of estimation, the influence of the changing p'_0 on the natural frequency is ignored, yielding $f = 8$ MHz, corresponding to a period of $0.125 \mu\text{s}$ for each fluctuation cycle. The actual periods of small fluctuations in Figs. 3(a) and 3(b) are estimated to be 0.124 and $0.125 \mu\text{s}$, which indeed correspond to the period of the bubble oscillation in the elastic region. The general analysis of this small amplitude fluctuation ends here. Although the period of the small fluctuations could be predicted, the bubble radius curve $R(t)$ always crosses the boundary line R_0 many times in one cycle, as shown in Figs. 2 and 3, making it difficult to be analytically solved. However, it is worth noting that although the vibration of the lipid bubble under low-frequency excitation is complex, we can still get the general trend, that is, compression at positive acoustic pressure and expansion at negative sound pressure. This feature can be used to reasonably estimate the vibration of a lipid bubble at low frequencies.

When the frequency of the incident wave is set higher than the resonant frequency, namely 7.0, 9.0 and 11 MHz, the vibration pattern of bubble radius is shown in Fig. 4. In order to clearly demonstrate the characteristics of bubble oscillation, the total duration of the three acoustic waves is set differently. When driven by high-frequency incident waves, the nonlinear behavior of the bubble is more complicated, with strong sub-harmonic oscillation appearing. Nevertheless, the fundamental component of bubble oscillation is still obvious. If we ignore the sub-harmonic vibration and only pay attention to the fundamental component which is the same as the excitation frequency, it can be seen in Fig. 4 that when the acoustic pressure rises, the bubble radius increases, and when the acoustic pressure decreases, the bubble gets smaller. This may seem counterintuitive, but it is actually consistent with the behavior of a damped mass-spring system. Please note that the

following conclusion may appear contrary to the observed phenomenon above. This is not an error but rather due to the negative sign preceding the excitation term p_{ac} in the bubble vibration equation. For a damped mass-spring system undergoing forced oscillation, it has a resonant frequency. On the one hand, when driven by a harmonic force with a frequency lower than the resonant frequency, the mass point oscillates in the “elastic control region,” where the amplitude of oscillation is largely controlled by the elasticity of the spring, and the phase is almost the same as the driving force. On the other hand, when driven by a harmonic force with a frequency higher than the resonant frequency, the mass point oscillates in the “mass control region,” where the amplitude of oscillation is mainly controlled by the mass, with a phase difference of π from the driving force. In the derivation below, the similarity of the damped mass-spring system and lipid bubble oscillation system will become more clear. Furthermore, the phase difference of π will be proved the root cause of negative ARF on lipid bubbles.

B. The analytical solution of the oscillation of lipid bubbles

Equation (1) is a nonlinear ordinary differential equation that describes the oscillation of a lipid-coated bubble. The independent variable R , the first derivative $\dot{R}$ and the second derivative $\ddot{R}$ are coupled with each other in the equation, with some of them even appearing as the denominator. Therefore, the equation should be linearized before further calculation.

When the lipid bubble oscillates under ultrasound excitation, the change of its radius is not too large, otherwise the bubble will rupture. The radius varies by no more than $1/10$, so the following assumptions are made:

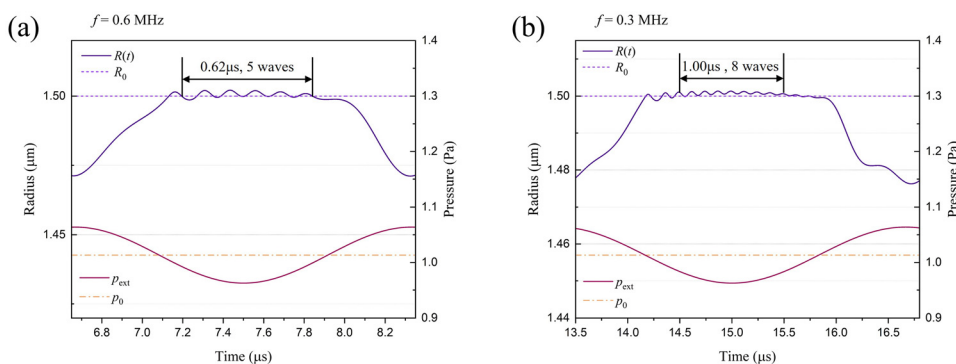


FIG. 3. The small amplitude fluctuation of the lipid coated bubble at low frequency during the negative acoustic pressure stage. The frequencies are (a) 0.6 and (b) 0.3 MHz.

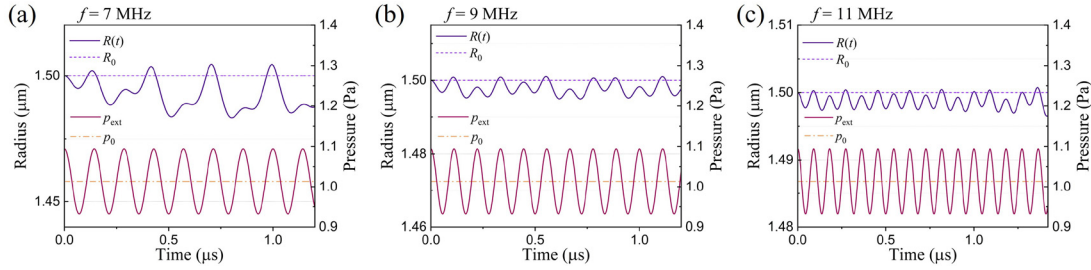


FIG. 4. The vibration curve of the lipid coated bubble under high-frequency excitation. The frequencies are (a) 7.0, (b) 9.0, and (c) 11 MHz.

$$R = R_0(1 + x), \quad (3)$$

where x is a small quantity relative to 1. Then, the first and second time derivatives of are $\dot{R} = R_0\dot{x}$ and $\ddot{R} = R_0\ddot{x}$. Substituting R in the equation with x , applying $\sigma(R_0) \approx 0$, and keeping at most quadratic terms of x yield

$$\begin{aligned} \rho_l R_0^2 \left[(1+x)\ddot{x} + \frac{3}{2}\dot{x}^2 \right] &= p_0 \left(1 - 3\kappa x + \frac{3\kappa(3\kappa+1)}{2}x^2 \right) \\ &\times \left(1 - R_0 \frac{3\kappa}{c} \dot{x} \right) - p_0 - \frac{2\sigma(x)}{R_0} (1-x+x^2) \\ &- 4\mu\dot{x}(1-x) - \frac{4\kappa_s}{R_0} \dot{x}(1-2x) - p_{ac}(t). \end{aligned} \quad (4)$$

Here, the following Taylor expansion is applied:

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2}x^2 + o(x^2). \quad (5)$$

In Eq. (4), $\sigma(x)$ is transformed from Eq. (2), where the ruptured case is omitted and the equation is linearized piecewise, since the bubble radius remains below $R_{break-up}$ in the case of small vibration. The piecewise linearized surface tension function is obtained by substituting $R_{buckling} \approx R_0$ and $R = R_0(1+x)$ into Eq. (2), and Taylor expanding it near $x = 0$, written as

$$\sigma(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ 2\chi x + o(1) & \text{if } x \geq 0 \end{cases} = 2\chi x \cdot u(x), \quad (6)$$

where $u(x)$ is the unit step function.

Combining Eqs. (6) and (4), keeping the primary terms of x in the equation, we get

$$\ddot{x} + \frac{1}{\rho_l R_0^2} \left(3p_0\kappa + \frac{4\chi \cdot u(x)}{R_0} \right) x + \frac{4}{\rho_l R_0^2} \left[\mu + \frac{\kappa_s}{R_0} - p_0 \frac{3\kappa R_0}{c} \right] \dot{x} = \xi(t), \quad (7)$$

where $\xi(t) = -\frac{p_a \cos \omega t}{\rho_l R_0^2}$ is the excitation term. Obviously, this is a piecewise ordinary differential equation, which is equivalent to

$$\begin{cases} \ddot{x} + \frac{3\kappa p_0}{\rho_l R_0^2} x + \frac{4}{\rho_l R_0^2} \left[\mu + \frac{\kappa_s}{R_0} - p_0 \frac{3\kappa R_0}{c} \right] \dot{x} = \xi(t) & x \leq 0 \\ \ddot{x} + \frac{3p_0\kappa + 4\chi/R_0}{\rho_l R_0^2} x + \frac{4}{\rho_l R_0^2} \left[\mu + \frac{\kappa_s}{R_0} - p_0 \frac{3\kappa R_0}{c} \right] \dot{x} = \xi(t) & x \geq 0. \end{cases} \quad (8)$$

The form of this equation implies that it is essentially a nonlinear damped mass-spring system with piecewise elasticity.

Let

$$\omega_1^2 = \frac{3p_0\kappa + 4\chi/R_0}{\rho_l R_0^2}, \quad (9)$$

which is the undamped intrinsic angular frequency of the lipid bubble when it oscillates in the high elasticity region ($x \geq 0$), and

$$\omega_2^2 = \frac{3\kappa p_0}{\rho_l R_0^2}, \quad (10)$$

which is the undamped intrinsic angular frequency of the lipid bubble when it oscillates in the low elasticity region ($x \leq 0$). Then the damping term is simply denoted as α ,

$$\alpha = \frac{2}{\rho_l R_0^2} \left[\mu + \frac{\kappa_s}{R_0} - p_0 \frac{3\kappa R_0}{c} \right], \quad (11)$$

and the amplitude of the excitation term written as ξ_0

$$\xi_0 = -\frac{p_a}{\rho_l R_0^2}. \quad (12)$$

Finally, Eq. (8) turns into

$$\begin{cases} \ddot{x} + \omega_1^2 x + 2\alpha\dot{x} = \xi_0 \cos \omega t, & x \geq 0 \\ \ddot{x} + \omega_2^2 x + 2\alpha\dot{x} = \xi_0 \cos \omega t, & x \leq 0. \end{cases} \quad (13a) \quad (13b)$$

This is apparently a piecewise linear ordinary differential equation, similar to a piecewise linear mass-spring system whose elasticity function is not symmetrical about $x = 0$. Chicurel-Uziel⁴⁶ derived the analytical solutions of several types of symmetrical piecewise linear strings. Shaw *et al.*⁴⁷ studied the asymmetrical piecewise-linear oscillator, and according to their study, such piecewise oscillation system, when its segment point is $x = 0$ and not driven by external forces, has a natural frequency of

$$\Omega_n = \frac{2\Omega_1\Omega_2}{\Omega_1 + \Omega_2}, \quad (14)$$

where $\Omega_1^2 = \omega_1^2 - \alpha^2$ and $\Omega_2^2 = \omega_2^2 - \alpha^2$ are the natural frequencies for $x \geq 0$ and $x \leq 0$ cases, respectively. Although Shaw *et al.* gave the natural frequency without external force, and also deduced the phase diagram of the nonlinear mass-spring system without damping and external force, they failed to derive any analytical results with damping and external force. Nevertheless, Eq. (14) can still be used as a reasonable estimate of the natural frequency of Eqs. (13a) and (13b).

Just as shown in Fig. 1, the solution of lipid bubble oscillation near the resonant frequency is regular, which can be simply divided into two parts: $x \geq 0$ and $x \leq 0$. Therefore, it can be assumed that the radius of the lipid bubble only crosses the segment point twice in one cycle. Meanwhile, only the steady-state solution is considered, because the transient response decays to nearly 0 and the bubble exhibits periodic vibration (the steady-state oscillation is usually attained within a few cycles). Near the resonant frequency, there should be $\omega_0 \approx \Omega_n$, where ω_0 represents the angular frequency of the driving acoustic wave. According to these assumptions and rules, a diagram that shows the characteristics of lipid bubble oscillation near the resonant frequency is drawn (see Fig. 5).

In Fig. 5, it is assumed that t_0 is the starting point of a vibration period of the bubble radius, which is not known temporarily but will be available in later calculation. Set the other two time points at t_1 and t_2 when bubble radius crosses the segment point. Obviously, these two time points are related by $t_2 = t_0 + T$, where T is the period corresponding to the frequency of the incident wave, and $T = \frac{2\pi}{\omega_0}$. There are two initial conditions, the initial radius and the initial velocity. The initial radius is $x(t_0) = 0$, which has been implied by setting t_0 as the starting point. The initial velocity is set to be v_0 here, that is: $\dot{x}(t_0) = v_0$, and the velocity at t_1 is set v_1 , which means $\dot{x}(t_1) = v_1$. By far all the variables have been set, and all of them will be solved in the following derivation.

In Fig. 5, the curve between t_0 and t_1 is governed by Eq. (13a), so its expression must be part of the solution of this equation; Similarly, the curve between t_1 and t_2 is part of the solution of Eq. (13b).

The solution of Eq. (13a) when $\dot{x}(t_0) = v_0, x(t_0) = 0$ is:

$$x_1(t; t_0, v_0) = e^{-\alpha(t-t_0)} [A_1 \cos(\Omega_1(t-t_0)) + B_1 \sin(\Omega_1(t-t_0))] + \gamma_1 \cos \omega_0 t + \delta_1 \sin \omega_0 t, \quad (15)$$

its changing rate being

$$\begin{aligned} \frac{dx_1(t; t_0, v_0)}{dt} = & e^{-\alpha(t-t_0)} [-A_1 \Omega_1 \sin(\Omega_1(t-t_0)) \\ & + B_1 \Omega_1 \cos(\Omega_1(t-t_0))] - \alpha e^{-\alpha(t-t_0)} \\ & \times [A_1 \cos(\Omega_1(t-t_0)) + B_1 \sin(\Omega_1(t-t_0))] \\ & - \gamma_1 \omega_0 \sin \omega_0 t + \delta_1 \omega_0 \cos \omega_0 t, \end{aligned} \quad (16)$$

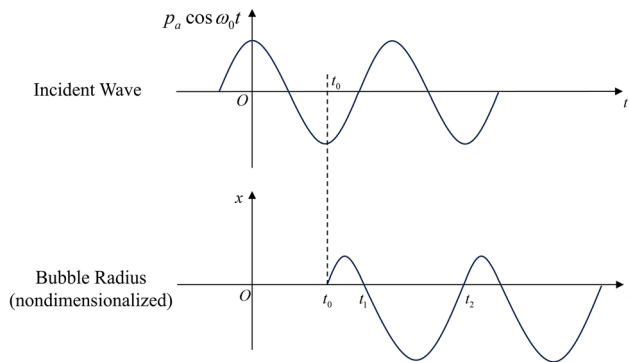


FIG. 5. Vibration diagram of the dimensionless radius of a bubble near the resonant frequency.

where the coefficients of the steady state terms are

$$\gamma_1 = \frac{\omega_1^2 - \omega_0^2}{(\omega_1^2 - \omega_0^2)^2 + (2\alpha\omega_0)^2} \xi_0, \quad (17)$$

$$\delta_1 = \frac{2\alpha\omega_0}{(\omega_1^2 - \omega_0^2)^2 + (2\alpha\omega_0)^2} \xi_0. \quad (18)$$

The coefficients of the transient terms are:

$$A_1 = -\gamma_1 \cos \omega_0 t_0 - \delta_1 \sin \omega_0 t_0, \quad (19)$$

$$B_1 = \frac{1}{\sqrt{\omega_1^2 - \alpha^2}} [v_0 + (\gamma_1 \omega_0 - \delta_1 \alpha) \sin \omega_0 t_0 - (\gamma_1 \alpha + \delta_1 \omega_0) \cos \omega_0 t_0]. \quad (20)$$

In a similar way, the solution of Eq. (13b) with the initial condition of $\dot{x}(t_1) = v_1, x(t_1) = 0$ is

$$x_2(t; t_1, v_1) = e^{-\alpha(t-t_1)} [A_2 \cos(\Omega_2(t-t_1)) + B_2 \sin(\Omega_2(t-t_1))] + \gamma_2 \cos \omega_0 t + \delta_2 \sin \omega_0 t, \quad (21)$$

its changing rate being

$$\begin{aligned} \frac{dx_2(t; t_1, v_1)}{dt} = & e^{-\alpha(t-t_1)} [-A_2 \Omega_2 \sin(\Omega_2(t-t_1)) \\ & + B_2 \Omega_2 \cos(\Omega_2(t-t_1))] - \alpha e^{-\alpha(t-t_1)} \\ & \times [A_2 \cos(\Omega_2(t-t_1)) + B_2 \sin(\Omega_2(t-t_1))] \\ & - \gamma_2 \omega_0 \sin \omega_0 t + \delta_2 \omega_0 \cos \omega_0 t, \end{aligned} \quad (22)$$

where the coefficients are

$$\gamma_2 = \frac{\omega_2^2 - \omega_0^2}{(\omega_2^2 - \omega_0^2)^2 + (2\alpha\omega_0)^2} \xi_0, \quad (23)$$

$$\delta_2 = \frac{2\alpha\omega_0}{(\omega_2^2 - \omega_0^2)^2 + (2\alpha\omega_0)^2} \xi_0, \quad (24)$$

$$A_2 = -\gamma_2 \cos \omega_0 t_1 - \delta_2 \sin \omega_0 t_1, \quad (25)$$

$$B_2 = \frac{1}{\sqrt{\omega_2^2 - \alpha^2}} [v_1 + (\gamma_2 \omega_0 - \delta_2 \alpha) \sin \omega_0 t_1 - (\gamma_2 \alpha + \delta_2 \omega_0) \cos \omega_0 t_1]. \quad (26)$$

So far, we have respectively found the solutions of piecewise linear ordinary differential equations (13a) and (13b) on the two segments $t_0 \sim t_1$ and $t_1 \sim t_2$. The next step is to connect the two solutions, Eqs. (15) and (21), with each other. There are four unknown variables to be solved, which are t_0, v_0, t_1 and v_1 . Using the boundary conditions of displacement continuity and velocity continuity at the segment point $x = 0$, four simultaneous equations are set up

$$\begin{cases} x_1(t_1, t_0, v_0) = 0 \\ \dot{x}_1(t_1, t_0, v_0) = v_1 \\ x_2(t_0 + T, t_1, v_1) = 0 \\ \dot{x}_2(t_0 + T, t_1, v_1) = v_0. \end{cases} \quad (27)$$

Solve the nonlinear simultaneous Eq. (27), substitute the results v_0 and v_1 into Eqs. (15) and (21), then divide and splice the solutions according to the results t_0 and t_1 , the final solution is obtained

TABLE II. A set of typical parameters for lipid coated bubbles.

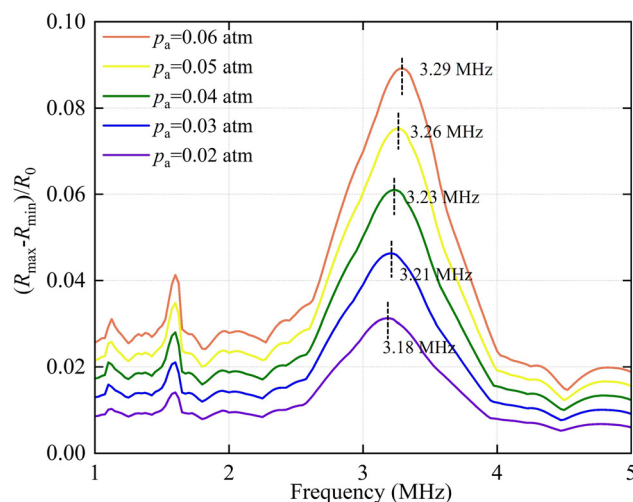
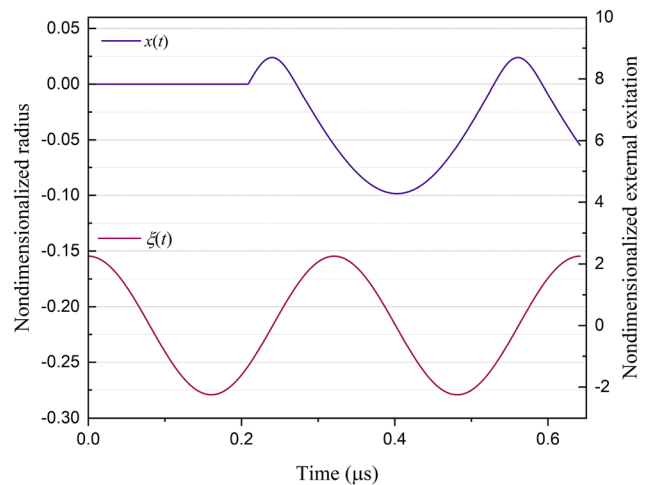
Parameters	Values
α	$8.93 \times 10^5 \text{ s}^{-1}$
ξ_0	$-2.25 \times 10^{12} \text{ s}^{-2}$
ω_1	$50.2 \times 10^6 \text{ s}^{-1}$
ω_2	$12.2 \times 10^6 \text{ s}^{-1}$
ω_0	$19.6 \times 10^6 \text{ s}^{-1}$

$$x(t) = \begin{cases} x_1(t; t_0, v_0), & t_0 + nT \leq t \leq t_1 + nT \\ x_2(t; t_1, v_1), & t_1 + nT \leq t \leq t_0 + (n+1)T. \end{cases} \quad (28)$$

Since the nonlinear simultaneous Eq. (27) do not have a universal solution, it is necessary to analyze each specific problem individually. A set of typical parameters for lipid bubbles are shown in Table II, which can be derived from Table I. The angular frequency ω_0 of the incident wave is intentionally set as the resonant frequency of the bubble using Eq. (14) (corresponding to 3.12 MHz), so that it can satisfy the applicable condition $\omega_0 \approx \Omega_n$. Before verifying the accuracy of the analytical solution, we first use numerical methods⁴⁴ to make sure that the formula (14) of natural frequency derived under the “no external force” condition is still applicable to give the resonant frequency of lipid bubbles driven by external ultrasound. The frequency response curve of a lipid bubble obtained by numerical methods is shown in Fig. 6. It can be found that the resonant frequency obtained by numerical methods is close to the theoretical frequency 3.12 MHz.

The data in Table II was substituted into the nonlinear Eq. (27), and the results are solved with the assistance of Matlab

$$\begin{cases} t_0 = 0.21 \mu\text{s} \\ t_1 = 0.27 \mu\text{s} \\ v_0 = 1.21 \mu\text{m/s} \\ v_1 = -1.18 \mu\text{m/s}. \end{cases} \quad (29)$$

**FIG. 6.** Frequency response curves of phospholipid coated bubbles at different incident wave amplitudes.**FIG. 7.** The vibration curve of the nondimensionalized bubble radius.

By substituting these results into Eq. (28), the vibration curve of the nondimensionalized bubble radius over one cycle during steady-state oscillation can be obtained, as shown in Fig. 7. The graph depicts the oscillation of $x(t)$ for $t > t_0$, alongside the external excitation curve. By comparing Fig. 7 with the numerical simulated result Fig. 1, we can see that, first, the “compression-only” behavior of the bubble is reproduced, indicating that the compression amplitude of the bubble radius is much greater than the expansion amplitude. Second, the amplitude of the bubble’s vibration indeed remains within 1/10 of its original radius, i.e., $x < 0.1$. Finally, the phase shift between the radius vibration and external excitation matches that of the numerical simulation.

In order to provide a clearer comparison, the numerical results of the bubble radius variation under the same conditions are also normalized, and then compared with the analytical solution, as shown in Fig. 8. The theoretical solution in the figure does not start from $t = 0$ but rather shifted 10 cycles to the right, commencing the comparison from the steady-state of oscillation. It can be observed that the theoretical solution matches the numerical computations. The alignment between the theoretical and numerical results include two aspects. For one thing, in terms of phase, there is nearly no phase difference between the two solutions, which is evident in the graph where the solid blue line and the dashed red line do not deviate horizontally. For another, regarding amplitude, there is only a slight discrepancy between the amplitudes of the two solutions, with the theoretical solution 6% larger than the numerical solution. As a result, the analytical solution for lipid bubble oscillation is validated as correct and feasible. Consequently, based on the analytical solution for bubble oscillation, it could be expected that the ARF on bubbles can be derived, which will be explored in subsequent sections.

Now let’s focus on the scenario where the frequency of incident acoustic waves is away from the resonant frequency.

As described earlier and illustrated in Fig. 2, the vibration of lipid bubble radius are complex at low frequencies, with unpredictable low-amplitude, high-frequency components. However, since the amplitude of the small fluctuations is small, it is reasonable to only consider the main trend of bubble oscillation. The nondimensionalized bubble radius is assumed to be

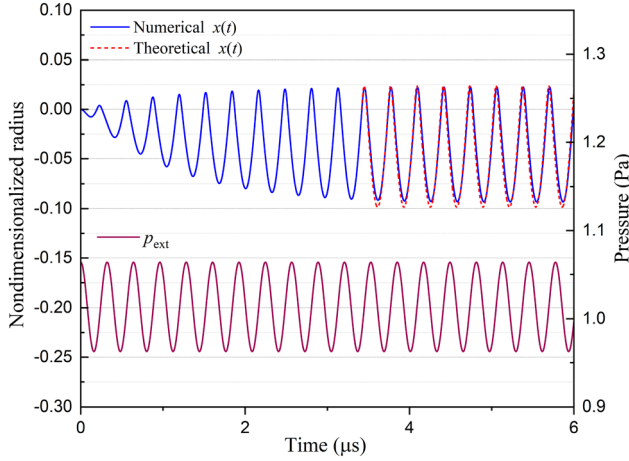


FIG. 8. Comparison between the numerical and analytical solutions of bubble oscillation.

$$x = C_1 e^{j(\omega_0 t + \phi_1)}, \quad (30)$$

so its first and second derivatives are $\dot{x} = C_1 j \omega_0 e^{j(\omega_0 t + \phi_1)}$ and $\ddot{x} = -C_1 \omega_0^2 e^{j(\omega_0 t + \phi_1)}$. Here, C_1 is an undetermined coefficient, and ϕ_1 is a phase constant. Substituting them into Eqs. (13a) and (13b), we get

$$-\omega_0^2 x + [\omega_2^2 + (\omega_1^2 - \omega_2^2)u(x)]x + 2j\alpha\omega_0 x = -\frac{p_a}{\rho_l R_0^2} e^{j\omega_0 t}. \quad (31)$$

When the excitation frequency is much lower than the resonant frequency, $\omega_0 \ll \omega_1, \omega_2$, the first term on the left side of the equation can be ignored. Then the piecewise equation turns into

$$[\omega_2^2 + (\omega_1^2 - \omega_2^2)u(x)]x + 2j\alpha\omega_0 x = -\frac{p_a}{\rho_l R_0^2} e^{j\omega_0 t}, \quad (32)$$

where the excitation term was previously defined in Eq. (12). Finally, the expression of nondimensionalized bubble radius for the low-frequency case is obtained

$$x = -\frac{\omega_2^2 + (\omega_1^2 - \omega_2^2)u(x) - 2j\alpha\omega_0 \frac{p_a e^{j\omega_0 t}}{\rho_l R_0^2}}{\omega_2^4 + (\omega_1^4 - \omega_2^4)u(x) + 4\alpha^2\omega_0^2 \frac{p_a e^{j\omega_0 t}}{\rho_l R_0^2}}. \quad (33)$$

Qualitative analysis on the above equation tells us that, when the damping term has minimal impact, i.e., $\alpha \ll \omega_1, \omega_2$, the bubble's radius has a phase shift of roughly π from the incident acoustic pressure. In other words, the bubble contracts when the acoustic pressure is positive, while expands when acoustic pressure is negative. The negative term with the imaginary unit j in the numerator of the equation implies that, apart from the π deviation, the bubble's phase will lag behind by another small value, which can be observed in Fig. 2.

The results calculated from Eq. (33) are compared with the numerically simulated bubble radius (see Fig. 9), and they match well with each other. The frequency chosen in the calculation is 0.3 MHz, which is much lower than the resonance frequency of the bubble. Although the theory directly neglected the high-frequency components and focused solely on low-frequency oscillation, with numerous approximations made during the derivation, the final results still successfully describe the general trend of bubble oscillation. This is particularly beneficial for the calculation of ARF in Sec. II C, as the ARF

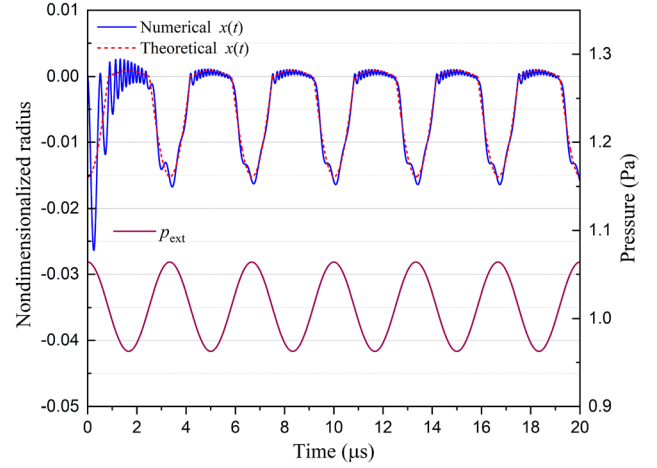


FIG. 9. Comparison of the theoretical result and numerical solution of the radius of a lipid bubble under low-frequency excitation.

is the time-average of the integral of the acoustic pressure on the particle surface. The components other than the fundamental mode have no effect on the time average calculation, as will be detailed in Sec. II C.

When exposed to high-frequency ultrasound, the lipid bubbles show complex sub-harmonic oscillation, making it harder to obtain its analytical expression. However, if we only focus on the fundamental components that have the same frequency as the excitation frequency, we can still solve the vibration of the bubble radius. Although such significant neglect may cause changes to the waveform of the radius curve, it is still acceptable as our aim is to calculate the ARF on lipid bubbles, and the components other than the fundamentals do not contribute to the ARF, which will be proved in Sec. II C.

The nondimensionalized bubble radius under high-frequency excitation is assumed as

$$x = C_2 e^{j(\omega_0 t + \phi_2)}, \quad (34)$$

so its first and second derivatives are $\dot{x} = C_2 j \omega_0 e^{j(\omega_0 t + \phi_2)}$ and $\ddot{x} = -C_2 \omega_0^2 e^{j(\omega_0 t + \phi_2)}$. Here, C_2 is an undetermined coefficient, and ϕ_2 is a phase constant. Substituting them into Eqs. (13a) and (13b) yields

$$-\omega_0^2 x + [\omega_2^2 + (\omega_1^2 - \omega_2^2)u(x)]x + 2j\alpha\omega_0 x = -\frac{p_a}{\rho_l R_0^2} e^{j\omega_0 t}. \quad (35)$$

When the excitation frequency is much higher than the resonant frequency, i.e., $\omega_0 \gg \omega_1, \omega_2$, the second term on the left side of the equation can be ignored. Then, the piecewise equation can be solved and the result is

$$x = \frac{\omega_0^2 + j2\alpha\omega_0 \frac{p_a e^{j\omega_0 t}}{\rho_l R_0^2}}{\omega_0^4 + 4\alpha^2\omega_0^2 \frac{p_a e^{j\omega_0 t}}{\rho_l R_0^2}}. \quad (36)$$

Suppose the damping term has minimal impact, i.e., $\alpha \ll \omega_1, \omega_2$, there should be almost no phase difference between the oscillation of bubble and its driving acoustic pressure, except for a small phase advance. In other words, when the acoustic pressure is positive, the bubble is getting bigger, and when the acoustic pressure is negative, the bubble is getting smaller.

Figure 10 shows the comparison of numerical results and theoretical values of bubble oscillation under high-frequency excitation (11 MHz). The direct numerical result in Fig. 10(a) is passed through a high-pass filter with a cutoff frequency of 10 MHz, so that only its fundamental component is retained and could be compared with the theoretical result in Fig. 10(b). Despite a small deviation in the amplitude of two curves, the theoretically calculated curve matches the numerical result in phase. Here, we focus solely on the comparison of the fundamental component, as the harmonic components do not affect the final calculation of the ARF. The ultimate goal of this work is to derive a simple and clear formula for calculating the ARF on lipid bubbles.

C. Theoretical derivation of acoustic radiation force on lipid bubbles

The original definition of ARF was the time average of the integral of acoustic pressure acting on the surface of a body in normal direction:⁴⁸

$$\langle F \rangle = - \left\langle \iint_S (P - p_0) \vec{n} \cdot d\vec{S} \right\rangle, \quad (37)$$

where $P = p_0 + p_{ac}$ is the total pressure outside the particle. The surface of a small particle is obviously a closed surface, so this formula can be converted to volume integral using Gaussian formula

$$\langle F \rangle = - \left\langle \iiint_V \nabla p_{ac} \cdot d\vec{V} \right\rangle. \quad (38)$$

Assuming that the radius of a particle, even when it is oscillating, is always far smaller than the wavelength of the incident wave, the pressure gradient ∇P can be approximated as a constant in the small volume V . Then, the integral above can be written as

$$\langle F \rangle = - \langle \nabla p_{ac} \cdot V \rangle, \quad (39)$$

which is exactly the definition of the Bjerknes force that can be applied on small oscillating particles.³⁴

It was difficult to obtain the ARF on lipid bubbles, as their radius vibration could only be obtained by numerical methods. Now it becomes possible since we have already got the analytical solution of the radius vibration of bubble, which could be easily converted to the change of bubble volume by

$$V(z, t) = \frac{4\pi R^3(z, t)}{3}, \quad (40)$$

where the radius is

$$R(z, t) = R_0(1 + x(z, t)). \quad (41)$$

A typical standing wave field is as follows:

$$p_{ac}(z, t) = p_a \sin kz \cos \omega t. \quad (42)$$

Substituting it into Eq. (1) as the excitation term, the ξ_0 in Eq. (13) becomes

$$\xi_0 = - \frac{p_a}{\rho_l R_0^2} \sin kz. \quad (43)$$

The gradient of the standing wave Eq. (42) is

$$\nabla p_{ac}(z, t) = k p_a \cos kz \cos \omega t. \quad (44)$$

Replacing the ∇p_{ac} and V in Eq. (39) with Eqs. (40) and (44), we obtain

$$\langle F \rangle = - \frac{4\pi k p_a \cos kz}{3} \langle R^3(z, t) \cos \omega t \rangle. \quad (45)$$

Here, R^3 can be expanded as

$$R^3(z, t) = R_0^3(1 + x)^3 = R_0^3(1 + 3x + 3x^2 + x^3). \quad (46)$$

According to the previous assumption, $x \sim o(1)$, only the first-order term needs to be retained, so the ARF on a lipid bubble, Eq. (45), can be converted into

$$\langle F \rangle = -4\pi k p_a R_0^3 \cos kz \cdot \frac{1}{T_0} \int_0^{T_0} x(z, t) \cos \omega_0 t \, dt, \quad (47)$$

where the time average symbol $\langle \cdot \rangle$ is translated into an integral, and T_0 means the period of $x(z, t) \cos \omega_0 t$, which depends on specific form of x . Now that the ARF formula has been simplified into Eq. (47), which is the time integral of nondimensionalized x multiplied by $\cos \omega_0 t$ over one vibration cycle, then multiplied by a constant. It is well known that the sine function has orthogonality, that is, the

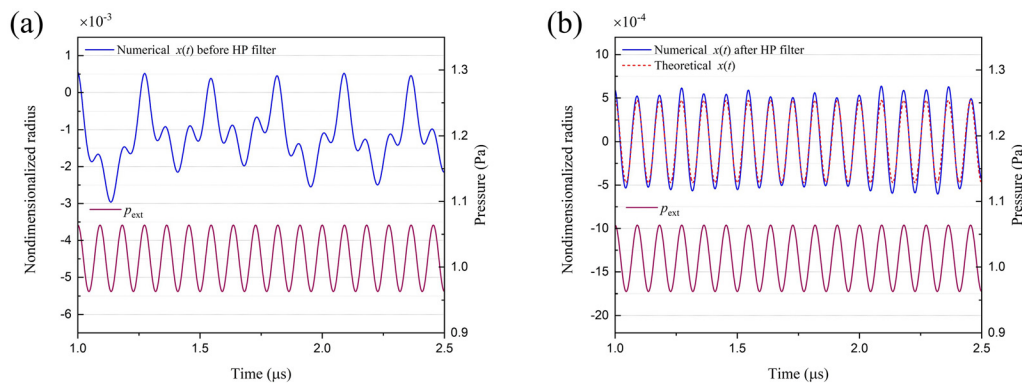


FIG. 10. Comparison of the theoretical result and numerical solution of the radius of a lipid bubble under high-frequency excitation (a) before high-pass filter and (b) after high-pass filter.

integral of the product of any two sinusoidal signals with different frequencies over one period is 0. Therefore, only when x is a sinusoidal function with a frequency of ω_0 , can Eq. (47) yield a non-zero value.

Along these lines, by expanding x using the Fourier series and retaining only the term with frequency ω_0 , we obtain

$$F = -2\pi k P_a R_0^3 \cos kz \cdot a_1, \quad (48)$$

which is the expression of the ARF on a lipid-coated bubble excited near the resonant frequency, and a_1 is the coefficient obtained from the Fourier series expansion

$$\begin{aligned} a_1 = & A_1 I(t_0, t_1, \Omega_1) + B_1 I(t_0, t_1, \Omega_1) + A_2 I(t_1, t_2, \Omega_2) + B_2 I(t_1, t_2, \Omega_2) \\ & + \frac{1}{2\omega_0 T} [(\gamma_1 - \gamma_2) \sin 2\omega_0 t_1 - \gamma_1 \sin 2\omega_0 t_0 - \gamma_2 \sin 2\omega_0 t_2] \\ & + \frac{\gamma_1 t_1 + \gamma_2 t_2 - (\gamma_1 + \gamma_2)t_0}{T} + \frac{1}{\omega_0 T} [\delta_2 \cos^2(\omega_0 t_2) \\ & - \delta_1 \cos^2(\omega_0 t_0) + (\delta_1 - \delta_2) \cos^2(\omega_0 t_1)], \end{aligned} \quad (49)$$

where I is an integral and its expression is

$$\begin{aligned} I(t_0, t_1, \Omega_1) = & \frac{2}{T} \int_{t_0}^{t_1} e^{-\alpha(t-t_0)} \cos \omega_0 t \cos(\Omega_1(t-t_0)) dt \\ = & \frac{\cos \Omega_1 t_0 e^{\alpha t_0}}{4T} \left[\begin{aligned} & \omega_0 (\sin(\omega_0 + \Omega_1)t_1 - \cos(\omega_0 + \Omega_1)t_1) \\ & e^{-\alpha t_1} \frac{-\alpha(\cos(\omega_0 + \Omega_1)t_1 + \sin(\omega_0 + \Omega_1)t_1)}{(\omega_0 + \Omega_1)^2 + \alpha^2} \\ & \omega_0 (\sin(\omega_0 + \Omega_1)t_0 - \cos(\omega_0 + \Omega_1)t_0) \\ & - e^{-\alpha t_0} \frac{-\alpha(\cos(\omega_0 + \Omega_1)t_0 + \sin(\omega_0 + \Omega_1)t_0)}{(\omega_0 + \Omega_1)^2 + \alpha^2} \\ & \omega_0 (\sin(\omega_0 - \Omega_1)t_1 + \cos(\omega_0 - \Omega_1)t_1) \\ & + e^{-\alpha t_1} \frac{-\alpha(\cos(\omega_0 - \Omega_1)t_1 - \sin(\omega_0 - \Omega_1)t_1)}{(\omega_0 - \Omega_1)^2 + \alpha^2} \\ & \omega_0 (\sin(\omega_0 - \Omega_1)t_0 + \cos(\omega_0 - \Omega_1)t_0) \\ & - e^{-\alpha t_0} \frac{-\alpha(\cos(\omega_0 - \Omega_1)t_0 - \sin(\omega_0 - \Omega_1)t_0)}{(\omega_0 - \Omega_1)^2 + \alpha^2} \end{aligned} \right]. \end{aligned} \quad (50)$$

Next, we consider the case of low-frequency excitation. Under conditions of low damping and low frequency, the imaginary part in equation Eq. (33) becomes so small that it can be neglected. Therefore, Eq. (33) can be easily transformed from a piecewise function segmented about x to a piecewise function segmented about time t ,

$$x = \begin{cases} -\frac{\omega_1^2}{\omega_1^4 + 4\alpha^2\omega_0^2} \frac{P_a}{\rho_l R_0^2} \cos \omega_0 t, & nT \leq t \leq \frac{T}{2} + nT \\ -\frac{\omega_2^2}{\omega_2^4 + 4\alpha^2\omega_0^2} \frac{P_a}{\rho_l R_0^2} \cos \omega_0 t, & \frac{T}{2} + nT \leq t \leq (n+1)T, \end{cases} \quad (51)$$

where n is an integer. Substituting Eqs. (42) and (51) into Eq. (47), we immediately obtain the ARF on a lipid bubble under low-frequency excitation

$$F = \frac{\pi k R_0 P_a^2}{2\rho_l} \sin 2kz \cdot \left[\frac{\omega_1^2}{\omega_1^4 + 4\alpha^2\omega_0^2} + \frac{\omega_2^2}{\omega_2^4 + 4\alpha^2\omega_0^2} \right]. \quad (52)$$

The case of high-frequency excitation is much easier. Substituting Eq. (36) into Eq. (47), yields

$$F = \frac{-\pi k R_0 P_a^2}{2\rho_l} \sin 2kz \cdot \frac{2\omega_0^2}{\omega_0^4 + 4\alpha^2\omega_0^2}. \quad (53)$$

This is the ARF on a lipid bubble under high-frequency excitation.

III. NUMERICAL SIMULATIONS

To validate the accuracy of the theory, numerical simulations are conducted with the help of MATLAB to analyze the ARF acting on lipid-coated bubbles.

The schematic diagram Fig. 11 illustrates the distribution of acoustic intensity for a standing wave. The points where acoustic intensity is zero correspond to nodes, and where it is maximum correspond to antinodes. At these points, either the pressure gradient or the pressure itself is consistently 0, hence, the ARF should be 0. Here, we select the midpoint between nodes and antinodes, precisely at the $1/8$ wavelength point of the sine function, as the position of the bubble in numerical simulations.

The bubble parameters used in the numerical simulations remain consistent with those listed in Table I. To reduce the influence of transient response on the results of ARF, the fourth-order Runge-Kutta method⁴⁵ is employed to determine the variation of the bubble radius $R(t)$ over 100 vibration cycles. Subsequently, only the results from the latter 50 cycles are utilized in Eq. (39) for computation. To validate the feasibility of the theory at different frequencies, the simulation frequency range is set from 0.1 to 80 MHz, so that the discrepancies could be observed between the numerical results and the theoretical formulations at low frequency Eq. (52), high frequency Eq. (53), and the resonant frequency Eq. (48).

Meanwhile, the results of our theory are also compared with the traditional Gorkov's formula. According to Henrik Bruus, the ARF on a tiny particle in the plane standing wave field is:³⁰

$$F_z^{rad} = -4\pi k R_0^3 \phi(\tilde{\kappa}, \tilde{\rho}) \cdot \frac{P_a^2}{4\rho_l c_l^2} \sin(2kz), \quad (54)$$

in which case the plane standing wave field is the same as Eq. (42). Here, ϕ is the acousto-phoretic contrast factor, and its expression is

$$\phi(\tilde{\kappa}, \tilde{\rho}) = \frac{1}{3} \left[\frac{5\tilde{\rho} - 2}{2\tilde{\rho} + 1} - \tilde{\kappa} \right], \quad (55)$$

where the $\tilde{\rho}$ and $\tilde{\kappa}$ are the density and compressibility ratios of between the particle and the medium, respectively. The compressibility of water here can be determined by $\kappa_l = \frac{1}{\rho_l c_l^2}$, and the gas compressibility is defined as $\kappa_p = -\frac{1}{V} \left(\frac{\partial V}{\partial p} \right)$, which can be easily derived by obtaining the partial derivative of ideal gas polytropic equation:⁴⁹

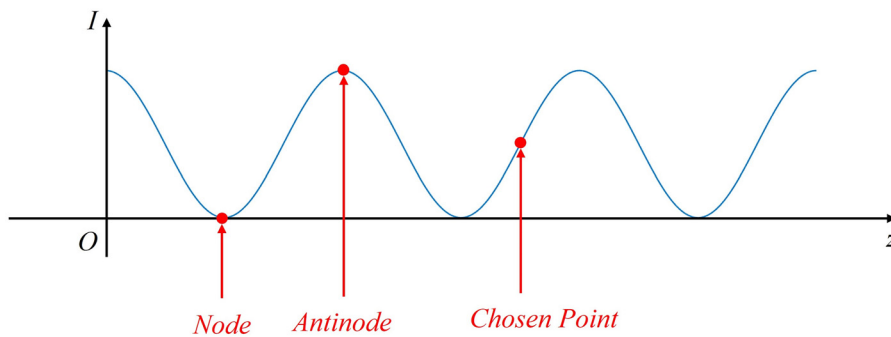


FIG. 11. The acoustic intensity distribution of acoustic standing wave and the selected point for numerical simulations.

$$p = \left[p_0 + \frac{2\sigma(R_0)}{R_0} \right] \left(\frac{R}{R_0} \right)^{-3\kappa}. \quad (56)$$

The gas in the bubble is sulfur hexafluoride with a density of 6.09 kg/m^3 .

The results are shown in Fig. 12.

First of all, the results confirm the speculation that the ARF changes with frequency. When the excitation frequency is lower than the resonant frequency, the ARF is positive, which means the bubble will move to the antinode where the acoustic pressure is higher. When the excitation frequency is higher than the resonant frequency, the ARF is negative and the bubble will move toward the node where the acoustic pressure is lower. This characteristic will be verified in the experiment.

Second, in Fig. 12, the red dashed line represents the results predicted by the low-frequency formula. Within the excitation frequency range of $f_0 < 1 \text{ MHz}$, the predicted results are consistent with the simulated results. This range is larger than the low-frequency approximation assumed in the previous theoretical derivation ($\omega_0 \ll \omega_1, \omega_2$, i.e., $f_0 \ll 1.93 \text{ MHz}$), suggesting the robustness of the new theory. As f_0

continues to increase and exceeds 1 MHz , the low-frequency approximation no longer holds. The ARF on the lipid bubble increases sharply, then suddenly decreases at the resonant frequency, becoming negative. On the other hand, the purple dash-dotted line draws the results predicted by the high-frequency formula. Within the excitation frequency range of $f_0 > 8 \text{ MHz}$, the predicted results match the numerical results. This exceeds the range specified by the high-frequency approximation assumption ($\omega_0 \gg \omega_1, \omega_2$, i.e., $f_0 \gg 7.99 \text{ MHz}$). The ARF formula near the resonant frequency Eq. (48) has more constraints. To calculate the bubble radius variation as described by Eq. (28), the bubble's oscillation must exhibit the regular "compression-only" mode shown in Fig. 1, which only occurs near the resonant frequency. Nevertheless, the predicted results of the ARF on a lipid bubble near the resonant frequency are consistent with numerical results in the frequency range from 2.9 to 4.0 MHz .

Third, the results of the Gorkov force, Eq. (54), are compared with the theoretical and numerical results, although the derivation of the Gorkov force determines that this formula is not applicable to bubbles with a density much lower than the medium and a compressibility much greater than the medium. As can be seen, only at low-frequency incident waves, far below the resonant frequency, does the Gorkov's formula approximate the theoretical value. At high frequencies, the Gorkov force still predicts a positive ARF which drags the bubble toward the high-pressure zone of the standing wave. Although the prediction of Gorkov's formula is consistent with its basic assumption: the direction of the ARF on the particle only depends on the acoustic contrast factor of the particle, previous experimental results proved that it is wrong,⁸ where the microbubbles driven above the resonant frequency are trapped at the pressure nodes.

IV. EXPERIMENTAL VERIFICATION

A. Experimental setup

In order to verify the correctness of the theory, the bubble aggregation behavior under different excitation frequencies is observed by experiments. A microfluidic experimental system is designed and built (see Fig. 13).

A water tank is placed on the microscope stage, and a microfluidic chip is fixed on the bottom of the water tank, with a plane wave ultrasound transmitting transducer and a solid aluminum block fixed on the left and right sides of the chip. The axis of the ultrasound transducer is perpendicular to the plane of the solid aluminum block, and the distance from the aluminum block is kept at an integer multiple of the emitted wavelength to obtain the best standing acoustic wave.

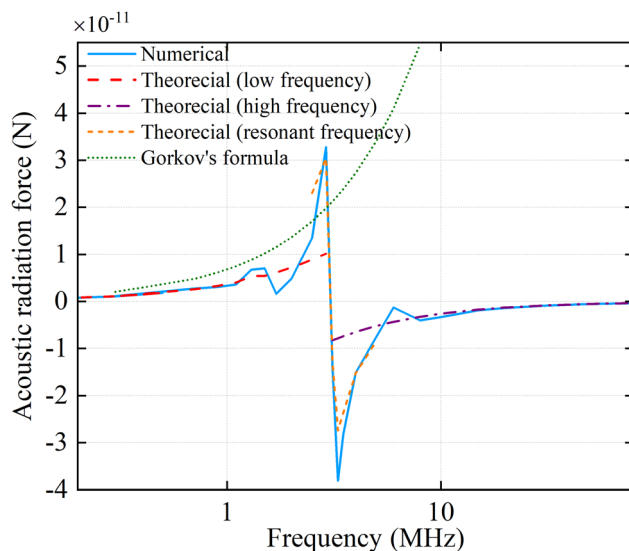


FIG. 12. The relationship between the ARF on a lipid bubble and driving frequency in a standing wave field.

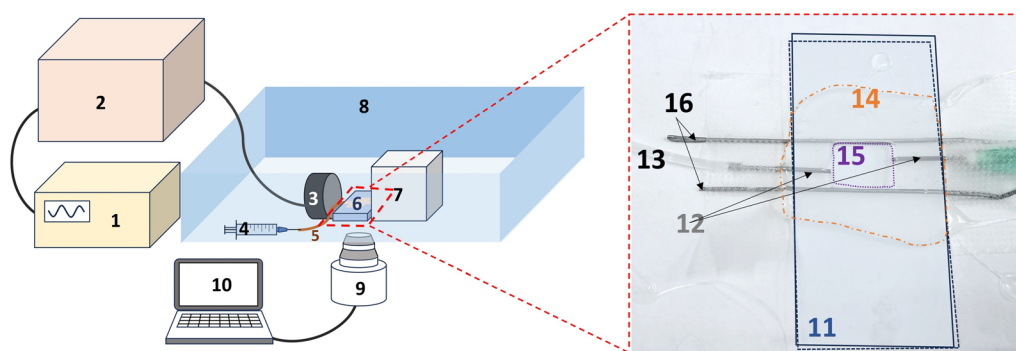


FIG. 13. Schematic diagram of the experiment system and microfluidic chip. 1—Signal generator, 2—Signal amplifier, 3—Ultrasound emitting transducer, 4—Syringe, 5—Plastic tubing, 6—Microfluidic chip, 7—Solid aluminum block, 8—Water tank, 9—Microscope, 10—Laptop, 11—Glass cover slips, 12—Hollow steel needles, 13—Plastic tubing (same as 5), 14—Hot melt adhesive, 15—Cavity, 16—Metal wires.

The field of microscope is centered on the microfluidic chip, and the experiment videos are recorded.

The core part of the microfluidic chip is a $1\text{ cm} \times 1\text{ cm} \times 1\text{ mm}$ flat cavity. The left and right sides of the cavity are respectively connected with liquid inlet and outlet. Its production method is as follows. Prepare two hollow steel needles, preheated hot melt adhesive gun, 1 mm diameter metal wire, medical tape, two glass cover slips, and place one coverslip on the white paper. Then the wire is cut into two sections with suitable length, straightened and placed on the coverslip so that the thickness of the cavity is controlled to be 1 mm, leaving a gap of 1 cm between the two wires. The two hollow steel needles are placed horizontally on the left and right sides of the metal cavity as the inlet and outlet of the liquid. After that, the steel needles and metal wires are stuck to the paper using medical tapes to prevent them from shifting during the next step. Afterward, the hot melt adhesive is squeezed onto the coverslip, and then another coverslip is quickly put on the melt adhesive, with tweezers pressing its surface so that the soft hot melt glue evenly fills the space around the cavity. Finally, after a few minutes, the hot melt adhesive cools down and successfully sticks the two cover slips together, leaving a suitable cavity for experiments.

Because the specific shape of the standing wave field cannot be directly observed under the microscope, polystyrene (PS for short) microspheres were used in the experiment to indicate the positions of the nodes in the standing wave field. PS microsphere, is a particle with its density slightly greater than water and compressibility slightly smaller than water, making them a typical example of “positive acoustic contrast factor” particles suitable for the Gorkov’s function. It has been widely used in numerous studies to visualize the shape of acoustic fields or as particles for ultrasound manipulation.^{50–53} These microspheres always aggregate at the points of minimum acoustic intensity in a gradient acoustic field. Therefore, in our experiments, they were used to determine the node positions of the standing wave field.

In the experiment, phospholipid-coated bubbles with an average diameter of $3.0\text{ }\mu\text{m}$ and PS microspheres with the same average diameter of $3.0\text{ }\mu\text{m}$ were used. Before starting the experiment, water mixed with lipid bubbles, and water mixed with PS microspheres, were drawn into two separate syringes. Another syringe filled with pure water was prepared for flushing particles out of the cavity. At the beginning of the experiment, the PS microsphere suspension was slowly injected into the plastic tubing to fill the cavity. At this point,

the signal generator, signal amplifier, and recording equipment were turned on. After the phenomenon stabilized, all devices were turned off, and water was injected into the cavity to flush out the PS microspheres. Then, the syringe containing the lipid bubble solution was connected, and the procedure was repeated while recording the experiment.

B. Results

First, we verified the ARF on lipid bubbles under low-frequency excitation. The movement and aggregation of PS microspheres and lipid bubbles were observed in a standing wave field generated by a 950 kHz plane wave ultrasound transducer, as shown in Fig. 14. Due to the differences in density and compressibility between the two types of particles, and the application of different input voltages, the motion velocities vary. The time points with the most significant phenomena from the experimental video were selected for analysis.

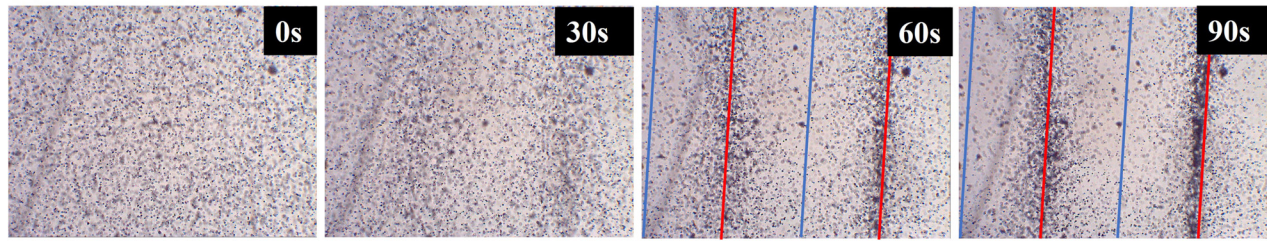
From the figure, it can be observed that after a period of movement, the PS microspheres gradually aggregate and form striped patterns, indicating the nodes of the standing wave field. These nodes are marked with red lines, while the antinodes, located at the midpoint between two nodes, are marked with blue lines. When the particles are replaced with phospholipid-coated bubbles, without altering the position of the ultrasound transducer, microfluidic chip, hard boundary, or microscope, the bubbles eventually gather at the antinode positions marked by the blue lines.

The bubbles do not form striped patterns like the PS microspheres because they are lightweight and subject to secondary Bjerknes forces,⁵⁴ which cause them to attract each other and form clusters when vibrating in phase. Therefore, the striped pattern is not stable for bubbles. Despite the interference of secondary Bjerknes forces, the bubble clusters still ultimately gather at the antinodes of the standing wave, rather than being repelled by nodes like the PS microspheres, which is consistent with theoretical predictions.

The same method was used to verify the ARF on lipid bubbles at high frequency. In the standing wave field generated by a 4 MHz plane wave ultrasound transducer, the motion of the two particles is shown in Fig. 15.

Similar to the low-frequency case, after a period of time, the PS microspheres aggregate at the nodes of the standing wave

(a) PS microspheres



(b) Lipid bubbles

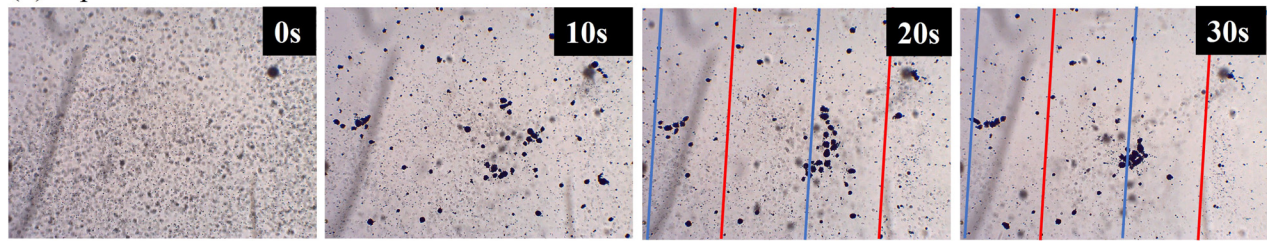
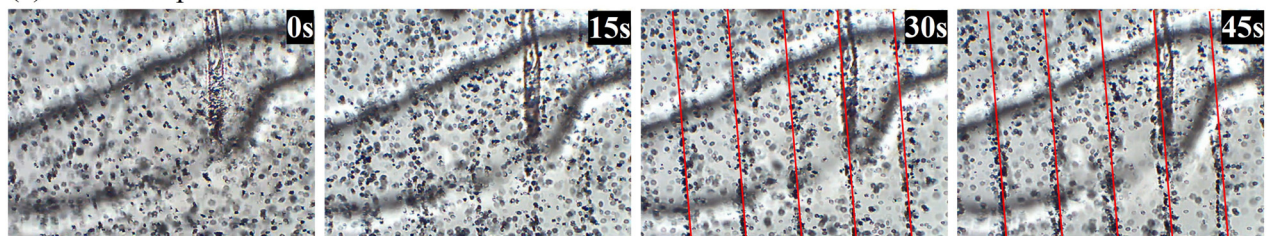


FIG. 14. The aggregation of (a) PS microspheres and (b) lipid bubbles driven by 950 kHz standing acoustic waves.

field, forming striped patterns. These aggregated areas are marked with red lines. After washing away the PS microspheres and replacing them with phospholipid-coated bubbles, it can be observed that the bubbles also slowly gather at the nodes, forming striped patterns at the red-marked lines. However, due to the influence of secondary Bjerknes forces between the bubbles, they attract each other to form clusters while aggregating into stripes, which may cause the striped pattern in Fig. 15(b) to seem discontinuous.

In the high-frequency standing wave field, the striped patterns formed by the bubbles are clearer than those at low frequencies. This could be attributed to two reasons. On the one hand, the high-frequency standing wave field has a greater lateral acoustic pressure gradient, resulting in a stronger ARF on the bubbles, which diminishes the relative influence of secondary Bjerknes forces. On the other hand, at high frequencies, bubbles aggregate at the nodes, where the acoustic pressure is at a minimum. In this region, the bubbles' vibrations are weaker (in a perfect standing wave field, bubbles at the nodal points

(a) PS microspheres



(b) Lipid bubbles

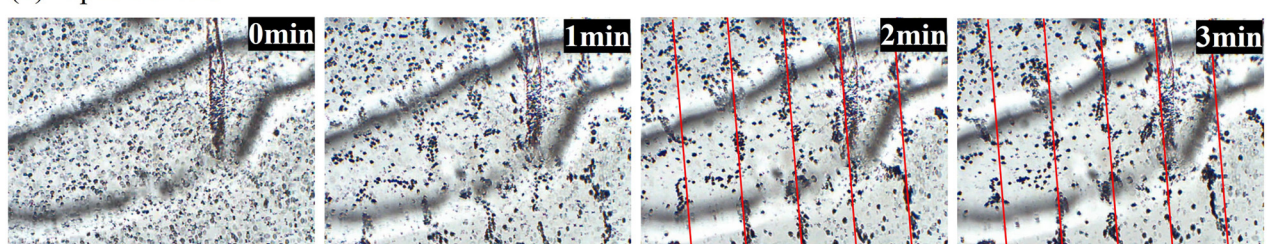


FIG. 15. The aggregation of (a) PS microspheres and (b) lipid bubbles driven by 4 MHz standing acoustic waves.

should not vibrate), resulting in minimal secondary Bjerknes forces between them. This reduces mutual attraction among bubbles, leading to clearer striped patterns.

V. CONCLUSION

In this work, a theoretical framework was proposed that first analytically calculates the radius variations of lipid-coated bubbles at different frequencies and then predicts the ARF acting on these bubbles. It was demonstrated that the direction of the ARF on lipid-coated bubbles is opposite under high-frequency and low-frequency excitation. The theoretical predictions were validated through numerical simulations and experiments. This theory complements the widely used Gorkov force, providing a better predictive framework for the forces acting on small particles with density and compressibility significantly different from the medium, addressing the limitation that Gorkov force is only applicable when the particle density and compressibility are similar to the medium. The proposed theory, especially Eqs. (48), (52), and (53), provides potential theoretical guidance for future studies on particle manipulation in biomedical ultrasound, particularly in the emerging area of manipulating gas-bearing biological particles.

VI. DISCUSSION

In this study, we observed completely opposite experimental results when the excitation frequency was below the bubble's resonant frequency compared to when it was above. These results align with theoretical predictions and are consistent with previous findings on free bubbles.^{26,37} However, previous researches on bubble manipulation have primarily utilized ultrasound transducers operating at frequencies above the resonance frequency, which causes bubbles to aggregate at the pressure nodes (weak acoustic pressure areas).^{8,55,56} Experiments using transducers with frequencies below the resonance frequency to control bubbles at pressure antinodes (strong sound field areas) are rare. This is likely due to the difficulty of achieving bubble aggregation in high-pressure regions for several reasons.

First and foremost, at lower frequencies, the acoustic pressure gradient is too small. Consequently, the ARF acting on bubbles is often smaller than the mutual attraction forces between bubbles (secondary Bjerknes forces). Therefore, in most cases, bubbles only form clusters due to secondary Bjerknes forces and do not move horizontally toward high-pressure regions. Second, using focused acoustic fields (with maximum acoustic pressure at the focal point) or vortex fields (with minimum acoustic pressure at the focal point) to manipulate bubbles can likely generate acoustic streaming. Light bubbles are easily carried away by the streaming. In a traveling wave, the maximum acoustic pressure coincides with the strongest acoustic streaming.^{57–59} Therefore, even if bubbles are attracted to this point, they are likely to be swept away by the streaming. This realization led us to use a standing wave field in our experiments. Third, if we attempt to input higher acoustic pressure to generate a stronger ARF, phospholipid-coated bubbles will rupture and aggregate into larger free bubbles.⁴⁴ These free bubbles have much larger radii and significantly lower resonant frequencies compared to phospholipid-coated bubbles. Consequently, they oscillate out of phase with the lipid bubbles, resulting in mutual repulsion due to secondary Bjerknes forces. This undoubtedly complicates the interactions between various types of bubbles, making it

more difficult to observe bubble aggregation in high-pressure regions. Fourth, bubbles, having a density lower than water, tend to float due to buoyancy and adhere to container ceilings. At low frequencies, the ARF is insufficient to overcome the drag force⁶⁰ and cause movement. Hence, bubbles are more likely to be driven by ARF on smooth surfaces like glass cover slips than in rough environments like plastic tubing. Fifth, another reason for using a standing wave field instead of a traveling wave focused field in our experiments is that, at the same frequency, the lateral acoustic pressure gradient in a standing wave field is larger, allowing for a stronger ARF on the bubbles. Although this factor is less significant compared to the previous points, it still contributes to the effectiveness of bubble manipulation in our experiments.

The experimental observations in this study serve as a qualitative verification of the theoretically predicted ARF reversal behavior. Due to several inherent limitations, a quantitative comparison between the experimental and numerical results was not made. The presence of multiple physical effects, such as microfluidic acoustic streaming and secondary Bjerknes forces, may influence bubble motion and complicate accurate estimation of ARF. These effects would need to be minimized in future experiments to enable precise force measurements. Moreover, although the bubble properties used in the numerical model were selected based on published literature, their consistency with the actual properties of the lipid-coated bubbles in our experiments remains uncertain. The frequency range explored was also limited. Near the resonance frequency, the bubbles often merge into larger bubbles due to a large amplitude of oscillation, whereas far from the resonant frequency, the ARF becomes too weak and may be overwhelmed by other effects. For these reasons, only two representative frequencies were selected to demonstrate the qualitative aggregation behavior. Future studies, if they are aimed at quantitative comparison, would benefit from using more stable bubbles and minimized the influence of other physical effects.

Although it is challenging to manipulate pure phospholipid-coated bubbles under low-frequency excitation, there are some similar experiments that can be explained or predicted using this theory. For example, gene editing techniques are used to generate bubbles within bacteria to make them easier to manipulate.¹² These bubbles are more stable than phospholipid-coated bubbles exposed to water. Due to the additional thick, elastic layer of the cell around the bubble, the resonant frequency may increase significantly, allowing them to be trapped at the focal point using a focused acoustic field. In future, researches on particle manipulation, various biological microparticles containing bubbles will emerge. Modeling these biological microparticles essentially involves adjusting the elasticity, damping, and inertia of the particles in our theory, as they are still fundamentally encapsulated bubbles. This theory can provide a theoretical foundation for the aggregation behavior of these microparticles in focused and vortex acoustic fields.

This theory is proposed for phospholipid-coated bubbles, but in essence, it should apply to any microbubble with a size much smaller than the wavelength. First, any microbubble with inertial and elastic properties will have a resonant frequency, analogous to a mass-spring system. When the excitation frequency is very low, the external acoustic pressure, if positive, will slowly compress the microbubble, reducing its radius. Therefore, the microbubble size is the smallest when the acoustic pressure is at its positive maximum and the largest when the

sound pressure is at its negative maximum. However, when the excitation frequency is very high, the influence of inertia far exceeds that of elasticity, causing the microbubble's radius to increase when the external acoustic pressure is positive. Consequently, the microbubble is largest when the sound pressure is at its positive maximum, which is to say, the microbubble is oscillating out of phase. This fundamentally determines that the direction of the ARF on the microbubble must be opposite when the excitation is above and below the resonant frequency.

Since every particle has a resonant frequency and the direction of the ARF changes accordingly, why is the Gorkov force formula correct in most cases? This is because, for a particle with density and compressibility close to that of the medium, its resonant frequency corresponds to a wavelength that is on the same order of magnitude as the particle size. Thus, Gorkov's "small particle assumption" falls within the low-frequency excitation range. In this range, the Gorkov force indeed aligns with the theory presented in this paper, as well as with the numerical simulations and numerous experimental studies. This insight suggests that in the manipulation of small particles, if the particle's density and compressibility are close to those of the medium, the Gorkov force can be used as a theoretical basis. However, if the particle's density and compressibility differ significantly from the medium, the Gorkov force becomes unreliable, and the theory presented in this paper offers a better predictive framework.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Xin Xu: Conceptualization (lead); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Software (lead); Supervision (lead); Validation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (equal). **Ye Yang:** Methodology (equal). **Longsheng Zeng:** Methodology (equal). **Minqin Shi:** Methodology (supporting). **Menyang Gong:** Project administration (equal); Writing – original draft (supporting); Writing – review & editing (equal). **Teng Ma:** Project administration (equal); Resources (equal). **Xiaozhou Liu:** Funding acquisition (lead); Project administration (lead); Resources (lead); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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